

OIS-006 — Place Runtime

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-005
 - OAS-004 — Place Architecture
 - OCS-000 through OCS-032
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Purpose

This specification defines the implementation responsibilities of the Place Runtime.

The Constitutional Specification defines what a Place is.

The Architectural Specification defines how a Place is organized.

This specification defines how a Place becomes an executable operational environment.

A Place Runtime is not an application.

A Place Runtime is the operational environment in which constitutional evolution occurs.

Every executable Place SHALL operate through a Place Runtime.

1. Implementation Objective

The objective of the Place Runtime is to provide an operational environment that faithfully realizes constitutional behavior.

The Runtime SHALL preserve:

Identity.

Memory.

Participation.

Conversations.

Events.

Learning.

Evolution.

Operational execution SHALL remain subordinate to constitutional behavior.

2. Runtime Lifecycle

Every Place Runtime SHALL support the following operational lifecycle.

Initialize



Load Constitution



Load Architecture



Load Schema



Restore Identity



Restore Event History



Construct Current State



Accept Participation

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Evolve Continuously ""

Every lifecycle transition SHALL remain observable.

3. Runtime Context

The Runtime SHALL maintain the constitutional context of the Place.

Context SHALL include:

Applicable Constitution.

Applicable Architecture.

Active Schema.

Canonical Identity.

Current Event Streams.

Operational projections.

Context SHALL remain internally consistent.

4. Runtime Services

The Runtime SHALL expose operational services for:

Identity Resolution.

Event Processing.

Conversation Management.

Library Access.

Decision Execution.

Learning Capture.

Federation Communication.

Reflection Support.

Services SHALL expose constitutional behavior.

5. Runtime State

Operational Runtime State SHALL be treated as disposable.

Runtime State SHALL be reconstructed from:

Events.

Schemas.

Identity.

Architectural relationships.

No Runtime State SHALL become constitutionally authoritative.

6. Runtime Recovery

Recovery SHALL reconstruct the complete operational environment using:

Canonical Identity.

Event history.

Schema Registry.

Architectural specifications.

Recovery SHALL preserve constitutional continuity.

7. Runtime Federation

Place Runtimes MAY participate in Federation.

Federation SHALL preserve:

Local sovereignty.

Canonical identity.

Event continuity.

Architectural compatibility.

Runtime Federation SHALL remain asynchronous.

8. Runtime Extensibility

Runtime capabilities MAY be extended through:

Additional Services.

Architectural Patterns.

Implementation modules.

Artificial Participants.

Extensions SHALL preserve constitutional compatibility.

9. Runtime Verification

Verification SHALL demonstrate:

Correct lifecycle execution.

Recovery correctness.

Identity continuity.

Behavioral equivalence.

Architectural compliance.

Operational explainability.

10. Runtime Pattern

The Place Runtime SHALL execute constitutional behavior.

Applications SHALL execute user experiences.

The Runtime SHALL remain independent of any specific application interface.

11. Place Runtime Test

A Place Runtime satisfies this specification when an independent engineering team can reconstruct an operational Place solely from its constitutional identity, architectural specifications, schemas, and preserved event history, producing constitutionally equivalent behavior without access to the original implementation.

Conformance

An implementation conforms to this specification when every Place executes within a Runtime that preserves constitutional identity, architectural organization, historical continuity, operational reconstructability, and implementation replaceability.

Implementation Principle

Applications come and go.

Interfaces evolve.

Technologies are replaced.

The Place remains.

The Runtime therefore exists not to execute software,

but to give constitutional life a stable operational environment in which understanding can continue to grow independently of the technologies that temporarily host it.

End of Specification OIS-006